

Chiyeong Heo

M.S. student • Graduate School of Artificial Intelligence, POSTECH • Machine Learning Lab
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Education

M.S. in Artificial Intelligence POSTECH, Graduate School of Artificial Intelligence • Pohang, South Korea Machine Learning Lab · Advisor: Prof. Jungseul Ok	2025 – Present
B.S. in Computer Science and Engineering Pusan National University • Busan, South Korea School of Computer Science and Engineering, College of Information and Bio-Medical Engineering Cumulative GPA 4.25 / 4.50 · Major GPA 4.30 Summa Cum Laude • President’s List	2020 – 2025

Experience

Research Intern, KAIST AI Research Internship (KAIRI) Cognitive Learning for Vision and Robotics (CLVR) Lab, KAIST • Seoul, South Korea Reinforcement learning research on reward design using Vision-Language Models (VLMs).	2024
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Publications

- Chiyeong Heo**, Jaechang Kim, Junhyuk Kwon, Hoyoung Kim, Dongmin Park, Jonghyun Lee, Jungseul Ok. “MMTB: Evaluating Terminal Agents on Multimedia-File Tasks.” *arXiv preprint*, 2026.
- Chi Yeong Heo**[†], Jun Seo[†], Yonggang Kim, Yohan Kim, Taewoon Kim. “Estimated Tardiness-Based Reinforcement Learning Solution to Repeatable Job-Shop Scheduling Problems.” *Processes*, 2025, 13(1), 62.

Patents

Method and Apparatus for Establishing Automated Production Planning Based on Factory Equipment and Job Status Inventors: Taewoon Kim, Jun Seo, Chiyeong Heo . Korean patent application (filed, under examination) • Assignee: Pusan National University Industry-Academic Cooperation Foundation Provisional: 10-2024-0192808 2024-12-20 • Main: 10-2025-0028136 2025-03-05

Awards & Honors

1st Place (Grand Prize), POSTECH Digital Twin Chatbot Hackathon Team project (DeepCatch Agent) with Jun Seo	2025
Academic Excellence Scholarship Pusan National University · merit-based	2023 – 2025
Busan Regional Talent Scholarship Busan Metropolitan City	2024 – 2025

Projects

DeepCatch Agent

1st place · 2025 POSTECH Digital Twin Chatbot Hackathon

Team project with Jun Seo (equal contribution)

An AI telephone-reservation assistant for fishing tourists in Pohang's Guryongpo: it gathers weather and catch data to recommend personalized fishing plans and automates phone bookings with fishing establishments through Twilio voice calls.

LangGraph · FastAPI · Twilio · OpenAI Realtime API · Socket.IO · Next.js / TypeScript

Mentoring

IBE Mentoring Program — Mentor

2024

Taught reinforcement-learning fundamentals to first- and second-year students: implemented and visualized policy and value iteration, and built a custom RL environment (Apple Game) with OpenAI Gymnasium.

Class Projects

Machine Learning & AI

- Project Einstein — An interactive physics-education platform that turns physics diagrams into animated simulations with AI-agent tutoring (computer vision + AI agents + real-time simulation). Team project with Jun Seo.
- Dong Baek Jeon Transaction Data Analysis [Data Mining](#) — K-means clustering of Busan local-currency transaction data to surface local economic patterns.
- Celebrity Look-Alike Finder [Cloud Computing](#) — An AI service using AWS Rekognition to find celebrity look-alikes; backend processing, API handling, and image ingestion.

Embedded Systems

- Human-Detecting Lifesaving 4WD Robot [Embedded System Design & Lab](#) — Firmware for a robot that detects people via PIR/ultrasonic sensors and communicates over Bluetooth (Cortex-M3 STM32F107, ESP32-CAM, L298N motor driver).

Software Development

- Zombie Survival FPS Game [Computer Graphics](#) — A 3D Unity game with NavMesh and rigidbody physics: player control, zombie AI, stage systems, and weapons.
- Hangman Android Game [Computer Software Design & Lab](#) — A Kotlin Android game with difficulty levels, sound effects, and a Firebase Firestore ranking system.
- Graduate School Lab Database [Database](#) — A contact-management and social web app with Next.js, Sequelize, and PostgreSQL; React UI and database transaction logic.
- Hangman Logic Circuit Design [Logic Circuit Design & Lab](#) — Hangman implemented as a digital circuit on an FPGA (Intel Quartus): keypad input, 7-segment display, and piezo output.